

Nicolas "Norswap" LAURENT

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Remote: ✓ / Relocate: ?

[PDF version](#)

Experience & Education

Founder & CEO

April 2024 - July 2025

Happy Devs

Happy Devs built HappyChain, a blockchain for fully-onchain games. Our main engineering focus was on building an onboarding experience for gamers unfamiliar with crypto, for which we built a state-of-the-art [embedded wallet and account abstraction stack](#), as well as some connext projects like chain monitoring infra and a novel onchain randomness system.

I ran the company, including fundraising, ops, managing a team of 6, and technical leadership.

Freelance Work & Explorations

February 2023 - March 2024

Wrote an exhaustive [research report](#) on cross-chain interoperability as the behest of the [Optimism Collective](#).

Built [roll-op](#), a deployment & devops tool for OP stack chains, which received multiple retro funding grants from the Optimism Collective.

Built [OxFable](#), a MVP fully-onchain trading card game (TCG), the first (and I think still only) of its kind. Notable for [innovative use](#) of zero-knowledge proofs to enforce random draws & keep players' hands hidden.

Delivered conferences talks at venues like EthDenver, ETH[CC] on onchain games and shared sequencing.

Staff R&D Protocol Engineer (Blockchain)

November 2021 - January 2023

Optimism, Remote

I worked on the design & architecture of the OP Stack, from inception to feature-completeness. The OP Stack is a L2 rollup blockchain stack, which currently secures [upwards of 20B\\$](#) in assets and more than a 1B\$ in daily transaction volume across more than a dozen blockchains.

I was involved in almost every design discussion, reviewed the entirety of the code base, and wrote the major part of the [specification](#). I also led discovery & cleanup work on Georges Hotz' initial implementation of [Cannon](#), the fault proof system responsible for securing the chain.

I dabbled in business development, where I was instrumental in bringing Velodrome Finance (a distributed exchange, now called Aero) onto OP Mainnet, where it held 100s of M\$ in value and achieved 10s of B\$ in cumulative trading volume. They multiplied these numbers one hundredfold on the Base blockchain (part of the Optimism Superchain) as Aerodrome Finance. I also helped handle the relationships with our "protocol partners", including organizations such as Flashbots, Shutter Network, Cartesi, ...

I delivered many conference talks, including at top venues like ETH[CC] and Devcon, on topics such as the OP Stack, Cannon, MEV-handling and cross-rollup atomic transactions.

Invited Professor

September 2020 - August 2022

Université Catholique de Louvain, Belgium

I taught the "*Languages & Translators*" compiler course to a class of about 100 computer science & engineering master students. Students had to implement their own programming language for the course's project.

The class lectures & materials are freely available at norswap.com/compilers

Core Developer Apprenticeship

June-October 2021

Ethereum Foundation, Remote

Senior R&D Engineer (Compilers)

October 2019 - November 2021

Oracle Labs, Remote

I worked on the GraalVM project, and in particular the implementation of TruffleRuby.

My work has contributed to improve TruffleRuby's peak performance (it is the fastest Ruby implementation on a wide range of benchmarks) and reduce its warmup time.

Much of my work at Oracle is publicly visible at github.com/oracle/truffleruby/commits?author=norswap

PhD in Computer Science

February 2014 - October 2019

Université Catholique de Louvain, Belgium

My thesis, "*Principled Stateful Parsing*" explores how adding custom code to parser specifications enables overcoming many practical challenges in the field, and how to mitigate the downsides.

Teaching assistant in the cloud computing, agile programming, and programming paradigms classes.

iOS Developer

September 2013 - February 2014

Famest, Louvain-La-Neuve, Belgium

Bachelor & Master in Computer Science

September 2008 - August 2013

Université Catholique de Louvain, Belgium

Grade: Magna Cum Laude

Double major in networking/security and software engineering.

Teaching assistant in the functional programming and computer systems classes.

Technologies

If you absolutely need to see a keyword in this section, we might not be a good fit. Having at this point implemented multiple languages and frameworks, I like to think I'm able to learn anything that is required in a reasonable amount of time.

That being said, I have experience in Java, C, Objective-C, Go, Kotlin, Ruby, Javascript, TypeScript, Python, Prolog, Lua, Bash, SQL, Git, Make, CMake, Maven, Gradle, Solidity, and more.

Selected Publications

Principled Procedural Parsing

Nicolas Laurent

PhD Thesis (2019); Louvain-La-Neuve, Belgium

Taming Context-Sensitive Languages with Principled Stateful Parsing

Nicolas Laurent and Kim Mens

ACM SIGPLAN Conference on Software Language Engineering (SLE) 2016; Amsterdam, The Netherlands

Parsing Expression Grammars Made Practical

Nicolas Laurent and Kim Mens

ACM SIGPLAN Conference on Software Language Engineering (SLE) 2015; Pittsburg, USA

SDLoad: An Extensible Framework for SDN Workload Generation.

Nicolas Laurent, Stefano Vissichio and Marco Canini

ACM SIGCOMM Workshop on Hot Topics in Software Defined Networking (HotSDN) 2014; Chicago, USA

A PEG-Based Macro System for Java

Nicolas Laurent

Master Thesis (2013); Louvain-La-Neuve, Belgium

A full list of publications, with download links, is available at norswap.com/publications

I also blog at norswap.com/sitemap on programming, blockchains, finance, exercise, and pop culture.

Spoken Languages

French

Native Speaker

English

Bilingual Proficiency (ILR-5)

Dutch

Limited Working Proficiency (ILR-2)